SYLLABUS & FORMAT OF OLYMPIAD

IEOM
INTERNATIONAL EDUCATION OLYMPIAD MATHEMATICS
MATHEMATICS

In general, the aim of the Olympiad in Mathematics is to assess the ability of the participant to apply in real life what he/she has learnt in school. The ultimate purpose is to identify talent and genius among the participants, identify the potential high achievers.

Therefore, Questions will be based on Understanding, Application, Logical Reasoning and Higher Order Mathematical Thinking Skills rather than rote learning. Evidently, the difficulty level will be above average.

CLASS 1 **(40 Marks)** **Time 1 Hour**	• **Shapes & Spatial Understanding (Geometry):** Vocabulary of Spatial Relationship (Top, Bottom, On, Under, Inside, Outside, Above, Below, Near, Far, Before, After); Solids Around Us; Sorting, Classifying and describing the objects on the basis of shapes, and other observable properties; Observing and describing the way shapes affect movements like rolling and sliding; Sorting 2 - D shapes. • **Numbers:** Developing a Sense of Numberless, Counting and Operations of numbers 1-9 and Zero; Arranging Collection of Objects in Order by Matching and One to One Correspondence; Counting Number of Objects in a Collection; Using Numbers from 1 to 9 in Counting and Comparison; Reading and Writing Numerals from 1 to 9; Adding and Subtracting (1 to 9) using Pictures; Adding and Subtracting the numbers using Symbols '+' and '-'; Approaches 'Zero' through the subtraction pattern (such as 3 – 1 = 2, 3 – 2 = 1, 3 – 3 = 0); Forms Number Sequence from 10 to 20, Counts Objects using these Numbers, Groups Objects into a Group of 10s and Single Objects, Developing the Vocabulary of Group of 'tens' and 'ones', Showing the group of tens and ones by drawing, Counting the number of tens and ones in a given number, Writing the numerals for eleven to nineteen, Writing numerals for ten and twenty, Comparing numbers up to 20, Addition and Subtraction (up to 20), Adding and subtracting numbers up to 20;
CLASS 2 **(40 Marks)** **Time 1 Hour**	• **Shapes & Spatial Understanding (Geometry):** 3-D and 2-D Shapes - Observing objects in the environment and getting a qualitative feel for their geometrical attributes, Identifying the basic 3-D shapes such as cuboid, cylinder, cone, sphere by their names, Tracing the 2-D outlines of 3-D objects, Observing and identifying 2-D shapes: viz., rectangle, square, triangle, circle by their names, Describing intuitively the properties of these 2-D shapes, Identifying and making straight lines by folding, straight edged objects, stretched strings and drawing free hand and with a ruler, Drawing horizontal, vertical and slant lines (free hand), Distinguishing between straight and curved lines

	• **Numbers:** Reading and Writing numerals for numbers up to ninetynine, Expanding a number with respect to place values, Counting and Regrouping objects into tens and ones, Using the concept of place value in the comparison of numbers, Counting in various ways: Starting from any number, Group Counting, etc., Arranging numbers up to hundred in ascending and descending order, Forming the greatest and the smallest two digit numbers with and without repetition of given digits, Indicating and identifying the position of an object in a line; Addition and Subtraction: Adding and subtracting two digit numbers by drawing representations of tens and ones without and with regrouping, Adding Zero to a number and subtracting zero from a number, Observing the commutative property of addition through patterns, Solving addition, subtraction problems presented through pictures and verbal description, Describing orally the situations that correspond to the given addition and subtraction facts, Estimating the result of addition and subtraction and comparing the result with another given number; Preparation for Multiplication and Division: Adding and subtracting single digit numbers mentally, Adding and subtracting multiples of ten mentally, Adds and subtracts small amounts of money mentally • **Measurement: Length -** Distinguishing between near, far, thin, thick, longer/taller, shorter, high, low, Seriating objects by comparing their length; Comparing between heavy and light objects; Time - Distinguishing between events occurring in time using terms - earlier and later
CLASS 3 **(50 Marks)** **Time 1 Hour**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West) • **Shapes & Spatial Understanding (Geometry):** Identifying 2-D shapes, Describing the various 2-D shapes by counting their sides, corners and diagonals, Making shapes on the dot-grid using straight lines and curves, Matching the properties of two 2-D shapes by observing their sides and corners (vertices), Reading simple maps (not necessarily scaled), Drawing some 3D-objects • **Numbers:** Reading and Writing 3-digit numbers, Expanding a number w.r.t. place values, Counting in different ways - starting from any number, Comparing numbers, Forms greatest and smallest numbers using given digits; Addition and Subtraction - Adding and subtracting numbers by writing them vertically – without regrouping and with regrouping, Using the place value in standard algorithm of addition and subtraction, Solving addition and subtraction problems in different situations presented through pictures and stories, Framing problems for addition and subtraction, Estimates the sum of, and difference between, two given numbers; Multiplication - Explaining the meaning of multiplication (as repeated addition), Identifies the sign of multiplication, Constructing the multiplication tables of 2, 3, 4, 5 and 10, Using multiplication facts in situations, Multiplying two digit numbers using standard algorithm and Lattice multiplication algorithm; Division - Explaining the meaning of division

	• from context of equal grouping and sharing, Relating division with multiplication, Completing division facts – by grouping or by using multiplication tables, Adding and subtracting single digit numbers and two digit numbers mentally, Doubling two digit numbers mentally (result not exceeding two digits); Money - Converting Rupee to Paise using play money, Adding and subtracting amounts using column addition, and subtraction without regrouping, Making rate charts and bills. • **Measurement: Length -** Measures lengths & distances along short & long paths using uniform (non-standard) units, extends to longer lengths, Weight - Comparing two or more objects by their weight, Capacity (Volume) – Comparing, and placing in order, containers in terms of internal volume (capacity), Placing in order given containers as per their capacities on the basis of perception & verifying by pouring out etc.; Time - Getting familiar with the days of the week and months of the year; Getting a feel for sequence of seasons, Sequencing the events occurring over longer periods in terms of dates/days .
CLASS 4 **(50 Marks)** **Time 1 Hour**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West). • **Shapes & Spatial Understanding (Geometry):** Drawing a circle free hand and with compass, Identifying centre, radius and diameter of a circle, Exploring intuitively the area and perimeter of simple shapes, Reading and drawing 3-D objects, making use of the familiarity with the conventions used in this, Drawing intuitively the plan, elevation and side view of simple objects. • **Numbers:** Numbers and Operations - Writing multiplication facts, Writing tables upto 10 × 10, Multiplying two and three digit numbers using lattice algorithm and the standard (column) algorithm; Dividing a given number by another number in various ways such as by drawing dots, by grouping, by using multiplication facts, by repeated subtraction; Applying the four operations to life situations, Framing Word Problems, Estimating sums, differences and products of given number; Adding and subtracting multiples of 10 and 100, mentally. Completes multiplication facts by adding partial products, mentally (e.g. 7 × 6 = 5 × 6 + 2 × 6); Fractional Numbers - Identifying half, one fourth and three- fourths of a whole, Identifying the symbols, 113 , , 244 . Explaining the meaning of 1 2 , 1 4 and 3 4, Appreciating equivalence of 2 4 and 1 2 ; and of 2 2 , 3 3 , 4 4 and 1; Money - Converting Rupees to Paise, Adding and subtracting amounts using column addition and subtraction with regrouping, Using operations to find totals, change, multiple costs and unit cost, Estimating roughly the totals and total cost; Identifying patterns in multiplication and division - multiples of 9, Casting out nines from a given number to check if it is a multiple of nine, Multiplying and Dividing by 10s, 100s, Identifying geometrical patterns based on symmetry

	• **Measurement:** Length **-**Appreciating the need for a standard unit, Measuring length using appropriate standard units of length by choosing between centimeters, and metres, Estimating the length of given object in standard units and verifying by measuring, Using a ruler, Relating centimetre. and metre; Weight – Weighing objects using non standard Units, Appreciating the conservation of weight; Volume - Measuring and comparing the capacity of different containers in terms of non-standard units; Time - Reading a calendar to find a particular day and date, Reading the time correct to the hour, Sequencing events chronologically**.**
CLASS 5 **(60 Marks)** **Time 1 Hours**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West) **.** • **Shapes & Spatial Understanding (Geometry):** Gets the feel of perspective while drawing a 3-D object in 2-D, Gets the feel of an angle through observation and paper folding, Identifies right angles in the environment, Classifies angles into right, acute and obtuse angles, Represents right angle, acute angle and obtuse angle by drawing and tracing; Explores intuitively rotations and reflections of familiar 2-D shapes, Explores intuitively symmetry in familiar 3-D shapes. • **Numbers:** Numbers and Operations - Finds place value in numbers beyond 1000, Appreciates the role of place value in addition, subtraction and multiplication algorithms, Uses informal and standard division algorithms, Explains the meaning of factors and multiples, Estimates sums, differences, products and quotients and verifies using approximation; Fractional Numbers - Finds the fractional part of a collection, Compares fractions, Identifies equivalent fractions, Estimates the degree of closeness of a fraction to known fractions (1/2, 1/4, 3/4 etc.), Uses decimal fractions in the context of units of length and money, Expresses a given fraction in decimal notation and vice versa; Money - Applies the four operations in solving problems involving money; Patterns - Identifies patterns in square numbers, triangular numbers, Relates sequences of odd numbers between consecutive square numbers, Makes border strip and tiling patterns**.** • **Measurement:** Length -Relating metre with centimeter, Converting metre into centimetres and vice versa, Solving problems involving length and distances, Estimating length of an object and distance between two given locations; Weight - Weighing objects using a balance and standard units, Determining sums and differences of weights, Estimating the weight of an object and verifying using a balance; Volume - Measuring volumes of given liquid using containers marked with standard units, Determining sums and differences of volumes, Estimating the volume of a liquid contained in a vessel and verifying by measuring; Time - Computing the number of weeks in a year, Correlating the number of days in a year with the number of days in each month, Justifying the reason for the need of a leap year, Reading clock time to the nearest hours and minutes, Expressing time, using the terms, 'a.m.' and 'p.m.', Estimating the duration of familiar events, Finding approximate time elapsed by (to the nearest hour) forward counting, Computing the number of days between two dates **n** **.**

CLASS 6 (60 Marks) Time 1 Hours	
	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics. **Numbers: (i) Knowing our Numbers:** Consolidating the sense of numberness up to 5 digits, Size, estimation of numbers, identifying smaller, larger, etc. Place value (recapitulation and extension), connectives: use of symbols =, and use of brackets, word problems on number operations involving large numbers up to a maximum of 5 digits in the answer after all operations. This would include conversions of units of length & mass (from the larger to the smaller units), estimation of outcome of number operations. Introduction to a sense of the largeness of, and initial familiarity with, large numbers up to 8 digits and approximation of large numbers) (ii) **Playing with Numbers:** Simplification of brackets, Multiples and factors, divisibility rule of 2, 3, 4, 5, 6, 8, 9, 10, 11. Even/odd and prime/composite numbers, Co-prime numbers, prime factorisation, every number can be written as products of prime factors. HCF and LCM, prime factorization and division method for HCF and LCM, the property LCM × HCF = product of two numbers. All this is to be embedded in contexts that bring out the significance and provide motivation to the child for learning these ideas. (iii) **Whole numbers, Natural numbers**, Properties of Numbers (commutative, associative, distributive, additive identity, multiplicative identity), Number Line. (iv) **Negative Numbers and Integers**: How negative numbers arise, models of negative numbers, connection to daily life, ordering of negative numbers, representation of negative numbers on number line. Children to see patterns, identify and formulate rules. What are integers, identification of integers on the number line, operation of addition and subtraction of integers, showing the operations on the number line (addition of negative integer reduces the value of the number) comparison of integers, ordering of integers (v) **Fractions**: Revision of what a fraction is, Fraction as a part of whole, Representation of fractions (pictorially and on number line), fraction as a division, proper, improper & mixed fractions, equivalent fractions, comparison of fractions, addition and subtraction of fractions (Large and complicated unnecessary tasks will be avoided); Abstraction in Fractions, Review of the idea of a decimal fraction, place value in the context of decimal fraction, inter conversion of fractions and decimal fractions (recurring decimals will be avoided at this stage), word problems involving addition and subtraction of decimals (two operations together on money, mass, length and temperature) . • **Algebra:** Introduction to Algebra - Introduction to variable through patterns and through appropriate word problems and generalisations (example 5 × 1 = 5 etc.), Generate such patterns with more examples; Introduction to unknowns through examples with simple contexts (single operations). • **Ratio and Proportion:** Concept of Ratio, Proportion as equality of two ratios, Unitary method (with only direct variation implied), Word problems . • **Geometry: (i) Basic Geometrical Ideas (2-D) -** Introduction to geometry; its linkage with and reflection in everyday experience, Line, Line Segment, Ray; Open and Closed Figures; Interior and Exterior of Closed Figures; Curvilinear and

	Linear Boundaries; Angle — Vertex, Arm, Interior and Exterior; Triangle — vertices, sides, angles, interior and exterior, altitude and median; Quadrilateral — Sides, Vertices, Angles, Diagonals, Adjacent Sides and Opposite Sides (only convex quadrilateral are to be discussed), interior and exterior of a quadrilateral; Circle — Centre, Radius, Diameter, Arc, Sector, Chord, Segment, Semicircle, Circumference, Interior and Exterior. (ii) **Understanding Elementary Shapes (2-D and 3-D)** - Measure of Line Segment, Measure of Angles, Pair of lines – Intersecting and Perpendicular Lines, Parallel lines; Types of Angles - Acute, Obtuse, Right, Straight, Reflex, Complete and Zero Angle; Classification of Triangles (on the basis of sides, and of angles); Types of Quadrilaterals – Trapezium, Parallelogram, Rectangle, Square, Rhombus; Simple Polygons (introduction) (Up to Octagons regulars as well as non regular); Identification of 3-D shape - Cubes, Cuboids, Cylinder, Sphere, Cone, Prism (triangular), Pyramid (triangular and square) - Identification and locating in the surroundings; Elements of 3-D figures. (Faces, Edges and Vertices); Nets for Cubes, Cuboids, Cylinders, Cones and Tetrahedrons. • **Measurement:** Determines area and perimeter of simple geometrical figures, Applies the four operations in solving problems involving length, weight and volume, Relates commonly used larger and smaller units of length, weight and volume and converts one to the other, Applies simple fractions to quantities, Converts fractional larger unit into complete smaller units, Appreciates volume of a solid body: intuitively and also by informal measurement, Uses addition and subtraction in finding time intervals in simple cases .
CLASS 7 **(60 Marks)** **Time 1 Hours**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics . • **Number System:** (i) Knowing our Numbers: Integers - Multiplication and division of integers (through patterns), Division by zero is meaningless, Properties of integers (including identities for addition & multiplication, commutative, associative, distributive) (through patterns) - These would include examples from whole numbers as well, Involve expressing commutative and associative properties in a general form, Construction of counterexamples, including some by children, Counter examples like subtraction is not commutative; Word problems including integers (all operations); (ii) Fractions and Rational numbers - Multiplication of fractions, Fraction as an operator, Reciprocal of a fraction, Division of fractions, Word problems involving mixed fractions; Introduction to Rational Numbers (with representation on number line), Operations on rational numbers (all operations), Representation of rational number as a decimal, Word problems on rational numbers (all operations), Multiplication and division of decimal fractions, Conversion of units (length & mass), Word problems (including all operations); (iii) Powers - Exponents only natural numbers, Laws of exponents (through observing patterns to arrive at generalisation (a) a to the power m x a to the power n = a to the power (m+n) (b) {a to the power m} to the power n = a to the power mn (c) a to the power n divided by a to the power of n = a to the power m-n where m-n Ɛ N .

	• **Algebra:** Algebraic Expressions, Generate algebraic expressions (simple) involving one or two variables, Identifying constants, coefficient, powers, Like and unlike terms, degree of expressions e.g., x y 2 etc. (exponent≤ 3, number of variables), Addition, subtraction of algebraic expressions (coefficients should be integers), Simple linear equations in one variable (in contextual problems) with two operations (avoid complicated coefficients). • **Ratio and Proportion:** Ratio and proportion (revision) Unitary method continued, consolidation, general expression; Percentage- an introduction - Understanding percentage as a fraction with denominator 100, Converting fractions and decimals into percentage and vice-versa, Application to profit and loss (single transaction only), Application to simple interest (time period in complete years). • **Geometry:** (i) Understanding shapes - Pairs of angles (linear, supplementary, complementary, adjacent, vertically opposite) (verification and simple proof of vertically opposite angles); Properties of parallel lines with transversal (alternate, corresponding, interior, exterior angles) (ii) Properties of triangles - Angle sum property (with notions of proof & verification through paper folding, proofs using property of parallel lines, difference between proof and verification), Exterior angle property, Pythagoras Theorem (Verification only) (iii) Symmetry - Recalling reflection symmetry Idea of rotational symmetry, observations of rotational symmetry of 2-D objects. (90, 120, 180 Degrees), Operation of rotation through 90 and 180 Degrees of simple figures, Examples of figures with both rotation and reflection symmetry (both operations), Examples of figures that have reflection and rotation symmetry and vice-versa; (iv) Representing 3-D in 2-D - Drawing 3-D figures in 2-D showing hidden faces, Identification and counting of vertices, edges, faces, nets (for cubes cuboids, and cylinders, cones), Matching pictures with objects (Identifying names), Mapping the space around approximately through visual estimation. • **Mensuration:** Concept of Perimeter and Introduction to Area, Introduction and General Understanding of Perimeter using many shapes, Shapes of different kinds with the same perimeter; Concept of area, Area of a rectangle and a square Counter examples to different misconcepts related to perimeter and area. Perimeter of a Rectangle, and its special case – a Square, Deducing the formula of the perimeter for a rectangle and then a square through pattern and generalisation . • **Data Handling: (i) What is Data -** choosing data to examine a hypothesis? **(ii) Collection and Organisation of Data -** examples of organising it in tally bars and a table. **(iii) Pictograph -** Need for scaling in pictographs, interpretation & construction. **(iv) Making Bar Graphs** for given data interpreting bar graphs.

CLASS 8 **(60 Marks)** **Time 1 Hours**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics . • **Number System:** (i) Rational Numbers - Properties of Rational Numbers. (including identities), Using general form of expression to describe properties, Consolidation of operations on rational numbers, Representation of rational numbers on the number line, Between any two rational numbers there lies another rational number (Making children see that if we take two rational numbers then unlike for whole numbers, in this case you can keep finding more and more numbers that lie between them.), Word problem (higher logic, two operations, including ideas like area); (ii) Powers - Integers as exponents, Laws of exponents with integral powers; (iii) Squares, Square roots, Cubes, Cube roots - Square and Square roots. Square roots using factor method and division method for numbers containing (a) no more than total 4 digits and (b) no more than 2 decimal places, Cubes and cubes roots (only factor method for numbers containing at most 3 digits), Estimating square roots and cube roots, Learning the process of moving nearer to the required number; (iv) Playing with numbers - Writing and understanding a 2 and 3 digit number in generalized form (100a + 10b + c , where a, b, c can be only digit 0-9) and engaging with various puzzles concerning this (Like finding the missing numerals represented by alphabets in sums involving any of the four operations.), Children to solve and create problems and puzzles, Number puzzles and games, Deducing the divisibility test rules of 2, 3, 5, 9, 10 for a two or three-digit number expressed in the general forms . • **Algebra:** (i) Algebraic Expressions - Multiplication and division of Algebraic Expressions (Coefficient should be integers), Some common errors (e.g. $2+x \neq 2x$, $7x+y \neq 7xy$), Identities (a±b)(square) = a(square) ± 2ab + b(square), a(square) – b(square) = (a–b) (a+b), Factorisation (simple cases only) as examples the following types a(x+y), (x±y)(square), a(square) – b(square), (x+a).(x+b), Solving linear equations in one variable in contextual problems involving multiplication and division (word problems) (avoid complex coefficient in the equations) . • **Ratio and Proportion:** Slightly advanced problems involving applications on percentages, profit & loss, overhead expenses, Discount, tax, Difference between simple and compound interest (compounded yearly up to 3 years or half-yearly up to 3 steps only), Arriving at the formula for compound interest through patterns and using it for simple problems, Direct variation – Simple and direct word problems, Inverse variation – Simple and direct word problems, Time & work problems– Simple and direct word problems . • **Geometry:** (i) Understanding shapes Congruence - Congruence through superposition (examplesblades, stamps, etc.), Extend congruence to simple geometrical shapes e.g. triangles, circles, Criteria of congruence (by verification) SSS, SAS, ASA, RHS; (vi) Construction (Using scale, protractor, compass), Construction of a line parallel to a given line from a point outside it

	(Simple proof as remark with the reasoning of alternate angles); Construction of simple triangles - Like given three sides, given a side and two angles on it, given two sides and the angle between them; Properties of Quadrilaterals – Sum of angles of a quadrilateral is equal to 360 Degrees (By verification), Properties of parallelogram (By verification) (a) Opposite sides of a parallelogram are equal, (b) Opposite angles of a parallelogram are equal, (c) Diagonals of a parallelogram bisect each other. [Why (d), (e) and (f) follow from (b)] (d) Diagonals of a rectangle are equal and bisect each other. (e) Diagonals of a rhombus bisect each other at right angles. (e) Diagonals of a square are equal and bisect each other at right angles;. • **Mensuration:** Revision of perimeter, Idea of Circumference of Circle, Area - Concept of measurement using a basic unit area of a square, rectangle, triangle, parallelogram and circle, area between two rectangles and two concentric circles. • **Data handling:** (i) Collection and organisation of data – choosing the data to collect for a hypothesis testing; (ii) Mean, median and mode of ungrouped data – understanding what they represent; (iii) Constructing bar graphs; (iv) Feel of probability using data through experiments, Notion of chance in events like tossing coins, dice etc., Tabulating and counting occurrences of 1 through 6 in a number of throws, Comparing the observation with that for a coin, Observing strings of throws, notion of randomness .
CLASS 9 **(60 Marks)** **Time 1 Hours**	• **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics . **1. Number Systems:** Real numbers. **2. Algebra –** Polynomials, Linear Equations in two Variables . **3. Coordinate Geometry:** The Cartesian plane, coordinates of a point, names and terms associated with the coordinate plane, notations, plotting points in the plane . **4. Geometry:** History - Geometry in India and Euclid's geometry; Lines and Angles; Triangles; Quadrilaterals; Area; Circles; Constructions . **5. Mensuration:** Revision of perimeter, Idea of Circumference of Circle, Area - Concept of measurement using a basic unit area of a square, rectangle, triangle, parallelogram and circle, area between two rectangles and two concentric circles . **6. Statistics:** (i) Collection and organisation of data – choosing the data to collect for a hypothesis testing; (ii) Mean, median and mode of ungrouped data – understanding what they represent; (iii) Constructing bar graphs .

CLASS 10 **(60 Marks)** **Time 1 Hour**	1. **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics. 2. **Number Systems:** Real numbers. 3. **Algebra:** Polynomials, Pair of Linear Equations in two Variables, Quadratic Equations, Arithmetic Progressions. 4. **Coordinate Geometry:** The Cartesian plane, coordinates of a point, names and terms associated with the coordinate plane, notations, plotting points in the plane, Lines (In two-dimensions). 5. **Trigonometry:** Introduction to trigonometry - Trigonometric ratios of an acute angle of a right-angled triangle, Proof of their existence (well defined); motivate the ratios whichever are defined at 0 and 90 Degrees, Values of the trigonometric ratios of 30, 45 and 60 Degrees, Relationships between the ratios; Trigonometric Identities, Proof and applications of the identity sin2A + cos2A = 1, Trigonometric ratios of complementary angles; Heights and Distances - Angle of elevation, Angle of Depression, Simple problems on heights and distances, not involve more than two right triangles, Angles of elevation/depression should be only 30°, 45°, and 60°. 6. **Geometry: History** - Geometry in India and Euclid's geometry; Lines and Angles; Triangles; Quadrilaterals; Area; Circles; Constructions. 7. **Mensuration:** Area of a triangle using Heron's formula (without proof) and its application in finding the area of a quadrilateral; Surface areas and volumes of cubes, cuboids, spheres (including hemispheres) and right circular cylinders/cones. 8. **Statistics & Probability:** Introduction to Statistics: Collection of data, presentation of data — tabular form, ungrouped / grouped, bar graphs, histograms (with varying base lengths), frequency polygons, Mean, Median and Mode of ungrouped data; Probability - History, Repeated experiments and observed frequency approach to probability, Focus on Empirical Probability .
CLASS 11 **(60 Marks)** **Time 1 Hour**	1. **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics. **2. Sets and Function:** Sets - Sets and their representations, Empty set, Finite and Infinite sets, Equal sets, Subsets, Subsets of a set of real numbers especially intervals (with notations), Power set, Universal set, Venn diagrams, Union and Intersection of sets, Difference of sets, Complement of a set, Properties of Complement; Relations & Functions - Ordered pairs, Cartesian product of sets, Number of elements in the Cartesian product of two finite sets,

	Cartesian product of the set of reals with itself (up to R x R x R), Definition of relation, Pictorial Diagrams, Domain, Co-domain and Range of a Relation, Function as a special type of relation, Pictorial representation of a function, domain, co-domain and range of a function. Real valued functions, domain and range of these functions, constant, identity, polynomial, rational, modulus, signum, exponential, logarithmic and greatest integer functions, with their graphs, Sum, difference, product and quotients of functions. 3. **Trigonometric Functions:** Positive and Negative Angles, Measuring angles in radians and in degrees and conversion from one measure to another, Definition of trigonometric functions with the help of unit circle, Truth of the identity sin2x + cos2x = 1, for all x, Signs of trigonometric functions, Domain and range of trigonometric functions and their graphs, Expressing sin (x±y) and cos (x±y) in terms of sinx, siny, cosx & cosy and their simple applications. Deducing identities involving sin, cos, tan, cot. 4. **Algebra:** Principle of Mathematical Induction; Complex Numbers and Quadratic Equations; Linear Inequalities; Permutations and Combinations; Binomial Theorem; Sequence and Series. 5. **Geometry & Co-ordinate :** History - Geometry in India and Euclid's geometry; Lines and Angles; Triangles; Quadrilaterals; Area; Circles; Constructions; Coordinate Geometry - The Cartesian plane, coordinates of a point, names and terms associated with the coordinate plane, notations, plotting points in the plane, Lines (In two-dimensions); Straight Lines; Conic Sections; Introduction to Three-dimensional Geometry. 6. **Statistics & Probability:** Introduction to Statistics: Collection of data, presentation of data — tabular form, ungrouped / grouped, bar graphs, histograms (with varying base lengths), frequency polygons, Mean, Median and Mode of ungrouped data; Probability - History, Repeated experiments and observed frequency approach to probability, Focus on Empirical Probability.
CLASS 12 **(60 Marks)** **Time 1 Hour**	1. **Mathematics:** What is Mathematics? Mathematics and Logic; Need and Importance of Mathematics; Origin and Development of Mathematics since ancient times (East and West); Branches of Mathematics . 2. **Relations and Functions:** Types of Relations - Reflexive, Symmetric, Transitive and Equivalence Relations; One to One and Onto Functions, Composite Functions, Inverse of a Function; Inverse Trigonometric Functions - Definition, Range, Domain, Principal Value Branch, Graphs of Inverse Trigonometric Functions, Elementary Properties of Inverse Trigonometric Functions.

	3. Algebra: Matrices; Determinants. **4. Calculus:** Limits and Derivatives; Limits and Derivatives Continuity and Differentiability; Applications of Derivatives; Integrals; Applications of the Integrals; Differential Equations. **5. Mathematical Reasoning:** Mathematically acceptable statements, Connecting words/ phrases - consolidating the understanding of "if and only if (necessary and sufficient) condition", "implies", "and/or", "implied by", "and", "or", "there exists" and their use through variety of examples related to real life and Mathematics, Validating the statements involving the connecting words, difference among contradiction, converse and contra positive. **6. Statistics and Probability:** Statistics - Measures of Dispersion: Range, Mean deviation, Variance and Standard Deviation of ungrouped/grouped data, Analysis of Frequency Distributions with equal means but different variances; Probability - Random experiments; outcomes, sample spaces (set representation). Events; occurrence of Events, 'not', 'and' and 'or' Events, Exhaustive Events, Mutually Exclusive Events, Axiomatic (set theoretic) Probability, Connections with other Theories, Probability of an event, probability of 'not', 'and' and 'or' events.